

Endangered Language Preservation with Machine Translation

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Introduction

Natural Language Processing (NLP) for Indigenous languages in the Americas, particularly the Muisca language from Colombia's Northeastern Andes, has been limited. The Muisca language, emerging around the 4th Century AD, has faced centuries of marginalization. Despite this, the Muisca community is actively revitalizing its culture and identity. We collaborated with the Muyscubun linguistics research group in Colombia to create the first parallel Muisca-Spanish corpus with 6,000 entries, developed from six unique documents of the Colección Mutis, housed in the Colombian Institute of Anthropology and History. Our creation of a Muisca corpus contributes to the preservation and revitalization of endangered languages, allowing future researchers to incorporate the newly created corpus in their linguistic research.

Background

The department of Boyacá, located in Colombia's Andean region, is rich in cultural elements deeply rooted in Muisca heritage. As such, the Muisca language, though extinct, has survived through colloquial use in the Spanish dialect spoken in Boyacá. An example of this is shown in Figure 1.

Furthermore, detailed records from Christian missionaries, who documented the language during the colonial period, have offered valuable insights into its grammatical structure. The Muisca community of Suba, Boyacá, Colombia are actively engaged in the process of re-indigenization by revitalizing their traditional knowledge, including working with the Muyscubun linguistics group, implementing Muisca in elementary schools, writing songs, and creating films in the language.

MUISCA	chahuana <i>adv free morpheme - Weak; no strength.</i> also spelled chajuana
COLOMBIAN SPANISH	Estoy achajuanado! I'm tired!

Figure 1: Muisca's influence on Colombian Spanish [1]

Methods

In this work we present, to the best of our knowledge, the first parallel Muisca-Spanish corpus. We leveraged a web-scraping library, Trafilatura, to extract raw text from the aforementioned data source. We then extracted both the Muisca phrases and their corresponding Spanish translation from Coleccion Mutis documents. We extract the Muisca words, which are surrounded by “***” when scraped because all Muisca text is in bold. The corresponding translations follow each phrase and are separated by a “=” character. Unfortunately, the inconsistent nature of these documents makes it difficult for automated web scraping tools to effectively gather and parse this data. Therefore, we combined our web scraper with manual inspection to create a cleaned corpus of 6,000 parallel sentences.

References

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Next Steps

We plan on training a neural machine translation system on our cleaned corpus by fine-tuning mT5[3], a multilingual variant of the T5 model [2].

To further support revitalisation efforts, we also plan to build an educational tool to teach Muisca to Spanish speakers. This tool will emphasize interactivity and natural conversation in order to provide an immersive learning experience. We also plan to integrate semantic similarity-based retrieval, and a Muisca to Spanish translator in our educational tool.

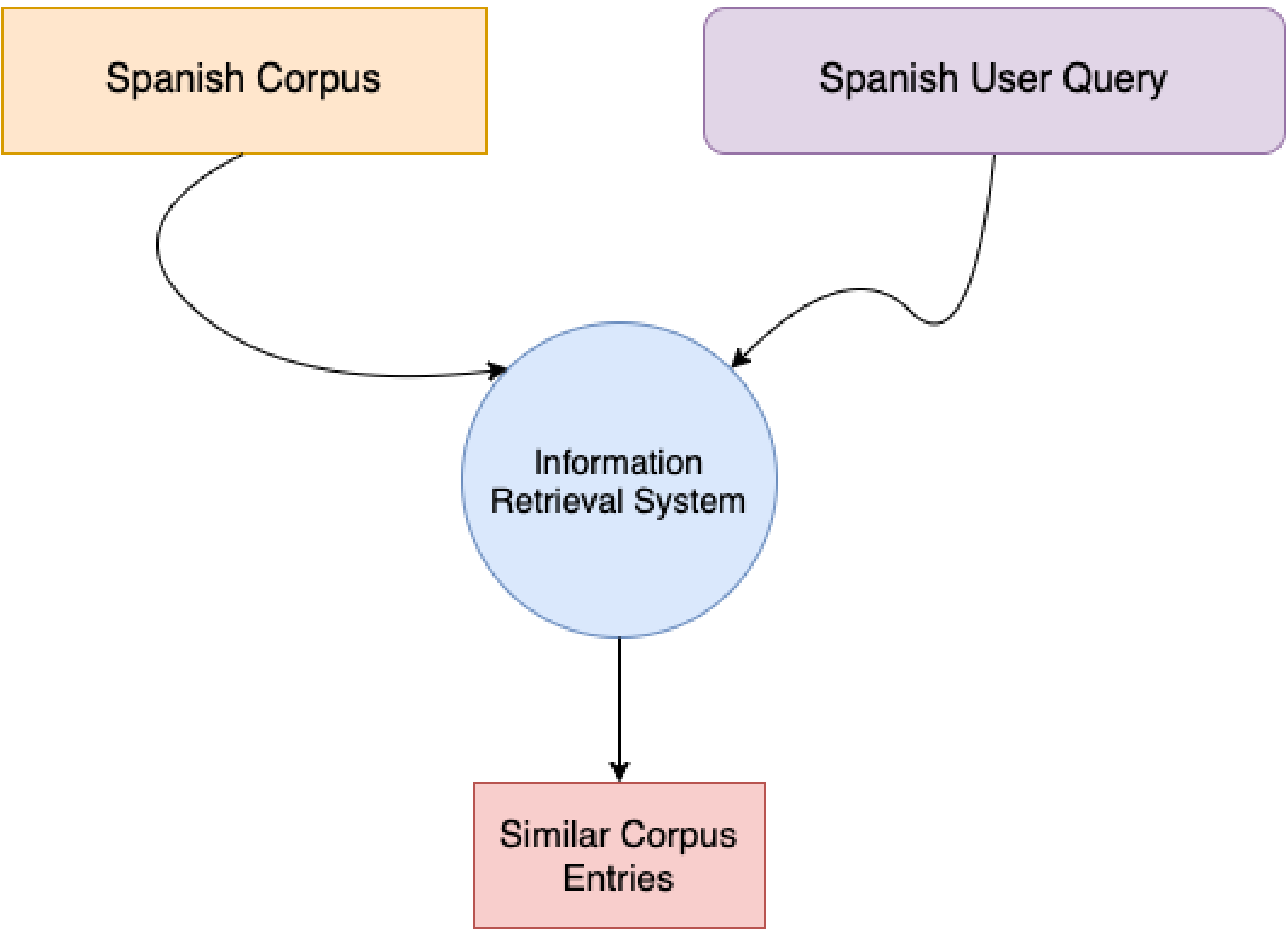


Figure 2: Proposed semantic-similarity pipeline

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